Proposed Solution and Architecture

Smart Irrigation Decision Agent

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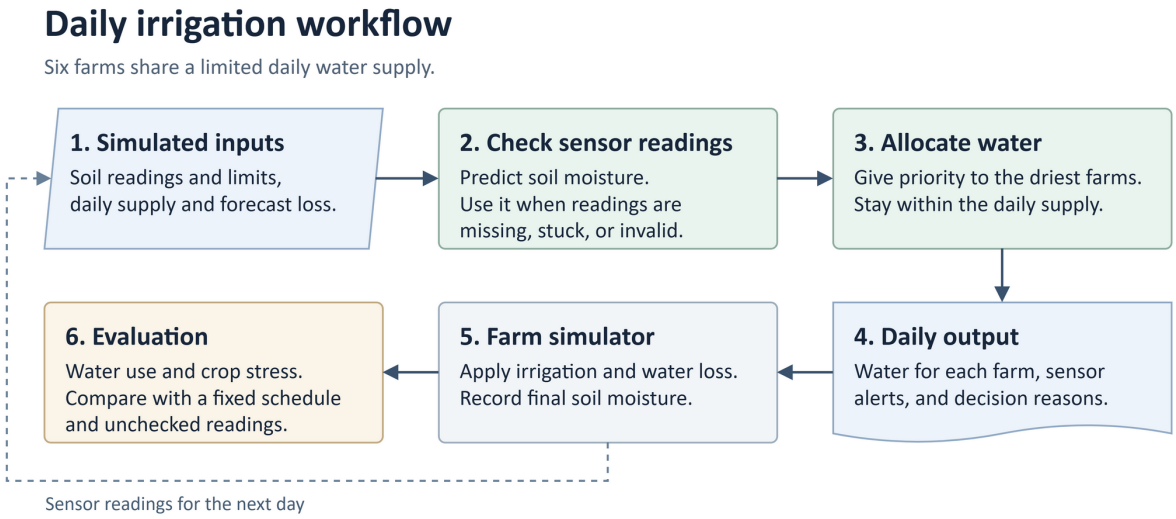


Figure 1. Daily irrigation workflow from draw.io. The rules and calculations run in the browser without an AI API or LLM.

AI pillars and daily decisions

Knowledge. The agent uses IF rules to reject a missing reading, a value outside the soil limits, a third identical reading, or a reading more than 10 mm from the prediction.

Uncertainty. The agent predicts moisture from the last estimate, water applied, and forecast loss. The check builds on [4] in Report 2. It uses the prediction for rejected readings and blends 60% of a passing reading with 40% of the prediction.

Optimization. The agent raises the lowest expected reserves first. It measures reserves in days of water above the crop stress level. In a shortage, **binary search** finds the highest reserve level the supply can cover. Each share is limited by soil capacity and the moisture target.

Tests and results

The five cases covered normal conditions, faulty readings, missing readings, 30% less water, and all problems together. Each strategy ran each case 20 times for 30 days. The agent, fixed schedule, and unchecked sensor strategy used the same starting soil, weather, and faults.

The agent flagged all 80 test faults within three days, raised 20 false alerts, and stayed within the water limit. In the combined test, the worst daily shortage averaged 2.54 days of normal water loss with the agent and 2.88 with the fixed schedule. The agent met the minimum moisture level on 18.2% of farm-days, compared with 20.8% for the fixed schedule.